

01).

i.

```
class Example {  
    public static void main(String[] args) {  
        int[] arr = {12, 35, 47, 59, 62, 73, 81, 93, 111, 121};  
    }  
}
```

ii.

Stack Memory		Heap Memory
arr	----->	[12, 35, 47, 59, 62, 73, 81, 93, 111, 121]
(ref)		Index: 0 1 2 3 4 5 6 7 8 9

iii.

Diagram (Before Assignment):

Heap Memory (Before)

[12, 35, 47, 59, 62, 73, 81, 93, 111, 121]
Index: 0 1 2 3 4 5 6 7 8 9

Diagram (After Assignment):

Heap Memory (After)

[99, 88, 47, 59, 62, 73, 81, 93, 111, 121]
Index: 0 1 2 3 4 5 6 7 8 9

iv.

```
class Example{  
    public static void main(String[] args) {  
        int[] arr = {12, 35, 47, 59, 62, 73, 81, 93, 111, 121};  
        System.out.println("arr[0]: " + arr[0]);  
        System.out.println("arr[3]: " + arr[3]);  
        System.out.println("arr[5]: " + arr[5]);  
    }  
}
```

```
}
```

02).

i.

```
class Example {  
    public static void main(String[] args) {  
        double[] averages = {85.66, 78, 67.33, 57.66, 81, 91, 53.66, 61.66, 74.33, 79.33};  
    }  
}
```

ii.

```
class Example {  
    public static void main(String[] args) {  
        double[] averages = {85.66, 78, 67.33, 57.66, 81, 91, 53.66, 61.66, 74.33, 79.33};  
        System.out.println("Average marks of the 3rd student: " + averages[2]);  
        System.out.println("Average marks of the 7th student: " + averages[6]);  
    }  
}
```

03).

i.

```
public class Example {  
    public static void main(String s[]) {  
        int[] reserves = {2362, 2311, 1917, 1912, 1887, 1854, 1717, 1717, 1779, 1705, 1806,  
1896};  
    }  
}
```

ii.

```
import java.util.Arrays;  
class Example{  
    public static void main(String[] args) {
```

```

    int[] reserves = {2362, 2311, 1917, 1912, 1887, 1854, 1717, 1717, 1779, 1705, 1806, 1896};

    System.out.println(Arrays.toString(reserves));

}
}

```

iii.

```

import java.util.*;

class Example {

    public static void main(String[] args) {

        int[] reserves = {2362, 2311, 1917, 1912, 1887, 1854, 1717, 1717, 1779, 1705, 1806, 1896};

        reserves[3] = 1812;

        reserves[6] = 1817;

    }

}

```

iv.

```

class Example{

    public static void main(String[] args) {

        int[] reserves = {2362, 2311, 1917, 1912, 1887, 1854, 1717, 1717, 1779, 1705, 1806, 1896};

        reserves[3] = 1812;

        reserves[6] = 1817;

        System.out.println("April - " + reserves[3]);

        System.out.println("July - " + reserves[6]);

    }

}

```

04).

i.

```

public class Example {

    public static void main(String s[]) {

```

```

        char[] spell =
{'l','n','s','t','i','t','u','t','e','O','f','C','o','m','p','u','t','e','r','E','n','g','i','n','e','e','r','i','n','g','T','e','c','h','n','o','l','o','g','y'};

        for(int i=0;i<spell.length;i++){

            System.out.print(spell[i]+" ");

        }

    }
}

```

ii.

```

public class Example {

    public static void main(String[] s) {

        char[] spell =
{'l','n','s','t','i','t','u','t','e','O','f','C','o','m','p','u','t','e','r','E','n','g','i','n','e','e','r','i','n','g','T','e','c','h','n','o','l','o','g','y'};

        String str = new String(spell);

        String[] words = str.split("(?=[A-Z])");

        System.out.println(String.join("\n", words));

    }

}

```

05).

i.

```

public class Example {

    public static void main(String s[]) {

        int[] marks = {56, 81, 43, 69, 93, 54, 48, 47, 51, 79, 82, 96, 57, 61, 66};

    }

}

```

ii.

marks.length

iii.

[56, 81, 43, 69, 93, 54, 48, 47, 51, 79, 82, 96, 57, 61, 66]

iv.

```
class Example {  
    public static void main(String[] args) {  
        int[] marks = {56, 81, 43, 69, 93, 54, 48, 47, 51, 79, 82, 96, 57, 61, 66};  
        for (int i = 0; i < marks.length; i++) {  
            System.out.println(marks[i]);  
        }  
    }  
}
```

v.

```
class Example {  
    public static void main(String[] args) {  
        int[] marks = {56, 81, 43, 69, 93, 54, 48, 47, 51, 79, 82, 96, 57, 61, 66};  
        for (int i = 9; i < marks.length; i++) {  
            System.out.println(marks[i]);  
        }  
    }  
}
```

06).

i.

```
public class Example {  
    public static void main(String s[]) {  
        String[] days =  
{"Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday", "Sunday"};  
        System.out.println(Arrays.toString(days));  
    }  
}
```

ii.

```

public class Example {

    public static void main(String s[]) {

        String[] days =
{"Monday","Tuesday","Wednesday","Thursday","Friday","Saturday","Sunday"};

        System.out.print("[");

        for(int i=days.length-1;i>=0;i--){

            System.out.print(days[i]+" , ");

        }

        System.out.print("\b\b]");

    }

}

```

07).

i.

```

public class Example {

    public static void main(String s[]) {

        char[] block =
{'A','B','C','D','E','F','G','H','I','J','K','L','M','N','O','P','Q','R','S','T','U','V','W','X','Y','Z'};

    }

}

```

ii.

```

public class Example {

    public static void main(String s[]) {

        char[] block =
{'A','B','C','D','E','F','G','H','I','J','K','L','M','N','O','P','Q','R','S','T','U','V','W','X','Y','Z'};

        System.out.println(Arrays.toString(block));

    }

}

```

iii.

```

public class Example {

```

```

public static void main(String s[]) {

    char[] block =
{'A','B','C','D','E','F','G','H','I','J','K','L','M','N','O','P','Q','R','S','T','U','V','W','X','Y','Z'};

    for(int i=0;i<block.length;i++){

        if(i%2==0){

            System.out.println(block[i]);

        }

    }

}

```

iv.

```

public class Example {

    public static void main(String s[]) {

        char[] block =
{'A','B','C','D','E','F','G','H','I','J','K','L','M','N','O','P','Q','R','S','T','U','V','W','X','Y','Z'};

        for(int i=0;i<block.length;i++){

            if(i%2!=0){

                System.out.println(block[i]);

            }

        }

    }

}

```

v.

```

public class Example {

    public static void main(String s[]) {

        char[] block =
{'A','B','C','D','E','F','G','H','I','J','K','L','M','N','O','P','Q','R','S','T','U','V','W','X','Y','Z'};

        for(int i=block.length-1;i>=0;i--){

```

```
System.out.println(block[i]);
```

```
}
```

```
}
```

```
}
```

08).

i.

```
public class Example {
```

```
    public static void main(String s[]) {
```

```
        int[] values = {2346, 1945, 2060, 1781, 2365, 1005, 1162, 1016, 1512, 2231, 1903,
12061, 1005, 1545, 2156, 2037, 1583, 3668};
```

```
    }
```

```
}
```

ii.

```
public class Example {
```

```
    public static void main(String s[]) {
```

```
        int[] values = {2346, 1945, 2060, 1781, 2365, 1005, 1162, 1016, 1512, 2231, 1903,
12061, 1005, 1545, 2156, 2037, 1583, 3668};
```

```
        int countDistrict = 0;
```

```
        for(int i=0;i<values.length;i++){
```

```
            countDistrict++;
```

```
        }
```

```
        System.out.println("Number of districts - "+countDistrict);
```

```
    }
```

```
}
```

iii.

```
public class Example {
```

```
    public static void main(String s[]) {
```



```

        int[] values = {2346, 1945, 2060, 1781, 2365, 1005, 1162, 1016, 1512, 2231, 1903,
12061, 1005, 1545, 2156, 2037, 1583, 3668};

        int count = 0;

        for(int i=0;i<values.length;i++){

            if(values[i]>2000){

                count++;

            }

        }

        System.out.println("Number of districts with more than 2000mm of annual rainfall -
"+count);

    }
}

```

iv.

```

public class Example {

    public static void main(String s[]) {

        int[] values = {2346, 1945, 2060, 1781, 2365, 1005, 1162, 1016, 1512, 2231, 1903,
12061, 1005, 1545, 2156, 2037, 1583, 3668};

        int count = 0;

        for(int i=0;i<values.length;i++){

            if(values[i]<1000){

                count++;

            }

        }

        System.out.println("Number of districts with less than 1000mm of annual rainfall -
"+count);

    }

}

```

v.

```

public class Example {

    public static void main(String s[]) {

```

```
int[] values = {2346, 1945, 2060, 1781, 2365, 1005, 1162, 1016, 1512, 2231, 1903,
12061, 1005, 1545, 2156, 2037, 1583, 3668};
```

```
int total = 0;
```

```
for(int i=0;i<values.length;i++){
```

```
    total+=values[i];
```

```
}
```

```
int avg = total/values.length;
```

```
System.out.println("Average annual rainfall values of all districts - "+avg);
```

```
}
```

```
}
```

09).

A,B,E,F

10).

i.

```
public class Example {
```

```
    public static void main(String s[]) {
```

```
        int[] rolling = {6,3,3,2,4,1,6,6,2,4};
```

```
    }
```

```
}
```

ii.

```
public class Example {
```

```
    public static void main(String s[]) {
```

```
        int[] rolling = {6,3,3,2,4,1,6,6,2,4};
```

```
        int count3 = 0;
```

```
        int count6 = 0;
```

```
        for(int i=0;i<rolling.length;i++){
```

```

        if(rolling[i]==3){
            count3++;
        }
    }
    for(int i=0;i<rolling.length;i++){
        if(rolling[i]==6){
            count6++;
        }
    }
    System.out.println("Number of times '3' - "+count3);
    System.out.println("Number of times '6' - "+count6);
}
}

```

iii.

```

public class Example {
    public static void main(String s[]) {
        int[] rolling = {6,3,3,2,4,1,6,6,2,4};
        int countOdd = 0;

        for(int i=0;i<rolling.length;i++){
            if(rolling[i]%2!=0){
                countOdd++;
            }
        }
        System.out.println("Number of odd numbers - "+countOdd);
    }
}

```

iv.

```

public class Example {

```

```

public static void main(String s[]) {
    int[] rolling = {6,3,3,2,4,1,6,6,2,4};
    boolean found = false;
    for(int i=0;i<rolling.length;i++){
        if(rolling[i]==5){
            found = true;
        }
    }
    if(found==true){
        System.out.println("Number 5 is rolled");
    }else{
        System.out.println("Number 5 is not rolled");
    }
}
}

```

11).

i.

//output→

0 0 0

int[] xr = new int[3]; creates an array of 3 integers, initialized to 0 by default. Line 4 (System.out.println(xr[0]+" "+xr[1]+" "+xr[2])); prints xr[0], xr[1], and xr[2], which are all 0, resulting in 0 0 0.

ii.

String array – null null null

Char array - not prints any characters

Boolean array – false false false

Double array – 0.0 0.0 0.0

12).

i.

```
import java.util.*;

public class Example {

    public static void main(String s[]) {

        Scanner input = new Scanner(System.in);

        int[] marks = new int[9];

        for(int i=0;i<marks.length;i++){

            System.out.print("Enter marks subject "+(i+1)+" - ");

            marks[i] = input.nextInt();

        }

        int total = 0;

        System.out.println(" ");

        for(int i=0;i<marks.length;i++){

            total+=marks[i];

        }

        System.out.println("Total marks of the student - "+total);

    }

}
```

ii.

```
import java.util.*;

public class Example {

    public static void main(String s[]) {

        Scanner input = new Scanner(System.in);

        int[] marks = new int[9];

        for(int i=0;i<marks.length;i++){

            System.out.print("Enter marks subject "+(i+1)+" - ");

            marks[i] = input.nextInt();

        }

        int total = 0;
```

```

        System.out.println(" ");
        for(int i=0;i<marks.length;i++){
            total+=marks[i];
        }
        System.out.println("Total marks of the student - "+total);
        int avg = 0;
        System.out.println(" ");
        for(int i=0;i<marks.length;i++){
            avg = total/marks.length;
        }
        System.out.println("Average marks of the student - "+avg);
    }
}

```

iii.

```

import java.util.*;

public class Example {
    public static void main(String s[]) {
        Scanner input = new Scanner(System.in);
        int[] marks = new int[9];
        for(int i=0;i<marks.length;i++){
            System.out.print("Enter marks subject "+(i+1)+" - ");
            marks[i] = input.nextInt();
        }
        int total = 0;
        System.out.println(" ");
        for(int i=0;i<marks.length;i++){
            total+=marks[i];
        }
        System.out.println("Total marks of the student - "+total);
    }
}

```

```

        int avg = 0;

        System.out.println(" ");

        for(int i=0;i<marks.length;i++){

            avg = total/marks.length;

        }

        System.out.println("Average marks of the student- "+avg);

        int mini = marks[0];

        System.out.println(" ");

        for(int i=0;i<marks.length;i++){

            if(marks[i]<mini){

                mini = marks[i];

            }

        }

        System.out.println("Minimum marks of the student - "+mini);

    }

}

```

iv.

```

import java.util.*;

public class Example {

    public static void main(String s[]) {

        Scanner input = new Scanner(System.in);

        int[] marks = new int[9];

        for(int i=0;i<marks.length;i++){

            System.out.print("Enter marks subject "+(i+1)+" - ");

            marks[i] = input.nextInt();

        }

        int total = 0;

        System.out.println(" ");

        for(int i=0;i<marks.length;i++){

```

```

        total+=marks[i];
    }
    System.out.println("Total marks of the student - "+total);
    int avg = 0;
    System.out.println(" ");
    for(int i=0;i<marks.length;i++){
        avg = total/marks.length;
    }
    System.out.println("Average marks of the student- "+avg);
    int mini = marks[0];
    System.out.println(" ");
    for(int i=0;i<marks.length;i++){
        if(marks[i]<mini){
            mini = marks[i];
        }
    }
    System.out.println("Minimum marks of the student - "+mini);
    int max = marks[0];
    System.out.println(" ");
    for(int i=0;i<marks.length;i++){
        if(marks[i]>max){
            max = marks[i];
        }
    }
    System.out.println("Maximum marks of the student - "+max);
}
}

```



```
import java.util.*;

public class Example {

    public static void main(String s[]) {

        Scanner input = new Scanner(System.in);

        int[] marks = new int[9];

        for(int i=0;i<marks.length;i++){

            System.out.print("Enter marks subject "+(i+1)+" - ");

            marks[i] = input.nextInt();

        }

        int total = 0;

        System.out.println(" ");

        for(int i=0;i<marks.length;i++){

            total+=marks[i];

        }

        System.out.println("Total marks of the student - "+total);

        int avg = 0;

        System.out.println(" ");

        for(int i=0;i<marks.length;i++){

            avg = total/marks.length;

        }

        System.out.println("Average marks of the student- "+avg);

        int mini = marks[0];

        System.out.println(" ");

        for(int i=0;i<marks.length;i++){

            if(marks[i]<mini){

                mini = marks[i];

            }

        }

        System.out.println("Minimum marks of the student - "+mini);

    }

}
```

```

        int max = marks[0];
        System.out.println(" ");
        for(int i=0;i<marks.length;i++){
            if(marks[i]>max){
                max = marks[i];
            }
        }
        System.out.println("Maximum marks of the student - "+max);
        int countGreater = 0;
        System.out.println(" ");
        for(int i=0;i<marks.length;i++){
            if(marks[i]>=75){
                countGreater++;
            }
        }
        System.out.println("The marks of greater than 75 - "+countGreater);
    }
}

```

vi.

```

import java.util.*;

public class Example {

    public static void main(String s[]) {

        Scanner input = new Scanner(System.in);
        int[] marks = new int[9];
        for(int i=0;i<marks.length;i++){
            System.out.print("Enter marks subject "+(i+1)+" - ");
            marks[i] = input.nextInt();
        }

        int total = 0;
    }
}

```

```
System.out.println(" ");
for(int i=0;i<marks.length;i++){
    total+=marks[i];
}
System.out.println("Total marks of the student - "+total);
int avg = 0;
System.out.println(" ");
for(int i=0;i<marks.length;i++){
    avg = total/marks.length;
}
System.out.println("Average marks of the student- "+avg);
int mini = marks[0];
System.out.println(" ");
for(int i=0;i<marks.length;i++){
    if(marks[i]<mini){
        mini = marks[i];
    }
}
System.out.println("Minimum marks of the student - "+mini);
int max = marks[0];
System.out.println(" ");
for(int i=0;i<marks.length;i++){
    if(marks[i]>max){
        max = marks[i];
    }
}
System.out.println("Maximum marks of the student - "+max);
int countGreater = 0;
System.out.println(" ");
```

```

        for(int i=0;i<marks.length;i++){
            if(marks[i]>=75){
                countGreater++;
            }
        }
        System.out.println("The marks of greater than 75 - "+countGreater);
        int countLess = 0;
        System.out.println(" ");
        for(int i=0;i<marks.length;i++){
            if(marks[i]<45){
                countLess++;
            }
        }
        System.out.println("The marks of less than 45 - "+countLess);
    }
}

```

vii.

```

import java.util.*;

public class Example {
    public static void main(String s[]) {
        Scanner input = new Scanner(System.in);
        int[] marks = new int[9];
        for(int i=0;i<marks.length;i++){
            System.out.print("Enter marks subject "+(i+1)+" - ");
            marks[i] = input.nextInt();
        }
        int total = 0;
        System.out.println(" ");
        for(int i=0;i<marks.length;i++){

```

```

        total+=marks[i];
    }
    System.out.println("Total marks of the student - "+total);
    int avg = 0;
    System.out.println(" ");
    for(int i=0;i<marks.length;i++){
        avg = total/marks.length;
    }
    System.out.println("Average marks of the student- "+avg);
    int mini = marks[0];
    System.out.println(" ");
    for(int i=0;i<marks.length;i++){
        if(marks[i]<mini){
            mini = marks[i];
        }
    }
    System.out.println("Minimum marks of the student - "+mini);
    int max = marks[0];
    System.out.println(" ");
    for(int i=0;i<marks.length;i++){
        if(marks[i]>max){
            max = marks[i];
        }
    }
    System.out.println("Maximum marks of the student - "+max);
    int countGreater = 0;
    System.out.println(" ");
    for(int i=0;i<marks.length;i++){
        if(marks[i]>=75){

```

```

        countGreater++;
    }
}
System.out.println("The marks of greater than 75 - "+countGreater);
int countLess = 0;
System.out.println(" ");
for(int i=0;i<marks.length;i++){
    if(marks[i]<45){
        countLess++;
    }
}
System.out.println("The marks of less than 45 - "+countLess);
boolean found = false;
System.out.println(" ");
for(int i=0;i<marks.length;i++){
    if(marks[i]==100){
        found = true;
    }
}
if(found==true){
    System.out.println("Is student have 100 marks in any subject - Yes");
}else{
    System.out.println("Is student have 100 marks in any subject - No");
}
}
}

```

13).

i.

Program A: Fills the array with user-entered integers using Scanner.

Program B: Fills the array with random integers using Random.

ii.

===== program B =====

```
import java.util.*;
class Demo{
    public static void main(String s[]) {
        Random r=new Random ();
        int[] numbers=new int[10];
        for (int i = 0; i < 10; i++){
            numbers[i]=r.nextInt(100)+1; // Line X
        }
        System.out.print("[");
        for (int i = 0; i < 10; i++){
            System.out.print(numbers[i]+", ");
        }
        System.out.println("\b\b");
    }
}
```

14).

i.

```
import java.util.*;
public class Example {
    public static void main(String s[]) {
        Random random = new Random();
        int[] marks = new int[20];

        for(int i=0;i<marks.length;i++){
            marks[i] = random.nextInt(100)+1;
        }
        System.out.println(Arrays.toString(marks));
        int min = marks[0];
        for(int i=0;i<marks.length;i++){
            if(marks[i]<min){
                min = marks[i];
            }
        }
    }
}
```

```

        }

    }

    System.out.println(" ");

    System.out.println("Minimum marks is - "+min);

}

```

ii.

```

import java.util.*;

public class Example {

    public static void main(String s[]) {

        Random random = new Random();

        int[] marks = new int[20];

        int countPass = 0;

        for(int i=0;i<marks.length;i++){

            marks[i] = random.nextInt(100)+1;

        }

        System.out.println(Arrays.toString(marks));

        for(int i=0;i<marks.length;i++){

            if(marks[i]>=50){

                countPass++;

            }

        }

        System.out.println(" ");

        System.out.println("The number of students pass the exam - "+countPass);

    }

}

```

iii.

```

import java.util.*;

public class Example {

```



```

public static void main(String s[]) {

    Random random = new Random();

    int[] marks = new int[20];

    int countPass = 0;

    for(int i=0;i<marks.length;i++){

        marks[i] = random.nextInt(100)+1;

    }

    System.out.println(Arrays.toString(marks));

    for(int i=0;i<marks.length;i++){

        if(marks[i]>=50){

            countPass++;

        }

    }

    boolean found = false;

    System.out.println(" ");

    System.out.println("The number of students pass the exam - "+countPass);

    for(int i=0;i<marks.length;i++){

        if(marks[i]<90){

            found = true;

        }

    }

    if(found==true){

        System.out.println("Is this have students of the marks more than 90 - Yes");

    }

    if(found!=true){

        System.out.println("Is this have students of the marks more than 90 - No");

    }

}

}

```

15).

i.

```
import java.util.*;

public class Example {

    public static void main(String s[]) {

        Scanner input = new Scanner(System.in);

        Random random = new Random();

        System.out.print("Enter array size - ");

        int size = input.nextInt();

        int[] arr = new int[size];

        for(int i=0;i<arr.length;i++){

            arr[i] = random.nextInt(100)+1;

        }

        System.out.println(Arrays.toString(arr));

    }

}
```

ii.

```
import java.util.*;

public class Example {

    public static void main(String s[]) {

        Scanner input = new Scanner(System.in);

        Random random = new Random();

        System.out.print("Enter array size - ");

        int size = input.nextInt();

        int[] arr = new int[size];

        int countOdd = 0;

        for(int i=0;i<arr.length;i++){

            arr[i] = random.nextInt(100)+1;
```

```

        if(i%2!=0){
            if(arr[i]%2!=0){
                countOdd++;
            }
        }
    }

    System.out.println(Arrays.toString(arr));
    System.out.println("Count odd numbers - "+countOdd);
}
}

```

iii.

```

import java.util.*;

public class Example {
    public static void main(String s[]) {
        Scanner input = new Scanner(System.in);
        Random random = new Random();
        System.out.print("Enter array size - ");
        int size = input.nextInt();
        int[] arr = new int[size];
        int countOdd = 0;
        for(int i=0;i<arr.length;i++){
            arr[i] = random.nextInt(100)+1;
            if(arr[i]==100){
                System.out.println("Found 100");
            }
        }

        System.out.println(Arrays.toString(arr));
    }
}

```

```
}
```

iv.

```
import java.util.*;
```

```
public class Example {
```

```
    public static void main(String s[]) {
```

```
        Scanner input = new Scanner(System.in);
```

```
        Random random = new Random();
```

```
        System.out.print("Enter array size - ");
```

```
        int size = input.nextInt();
```

```
        int[] arr = new int[size];
```

```
        for(int i=0;i<arr.length;i++){
```

```
            arr[i] = random.nextInt(100)+1;
```

```
        }
```

```
        int max = arr[0];
```

```
        int min = arr[0];
```

```
        System.out.println(Arrays.toString(arr));
```

```
        for(int i=0;i<arr.length;i++){
```

```
            if(arr[i]>max){
```

```
                max = arr[i];
```

```
            }
```

```
        }
```

```
        System.out.println("Maximum value of the array - "+max);
```

```
        for(int i=0;i<arr.length;i++){
```

```
            if(arr[i]<min){
```

```
                min = arr[i];
```

```
            }
```

```
        }
```

```
        System.out.println("Minimum value of the array - "+min);
```

```
}  
}
```

16).

i.

Length : 7

Element at index 0 is 10

Element at index 1 is 20

Element at index 2 is 30

Element at index 3 is 40

Element at index 4 is 50

Element at index 5 is 60

Element at index 6 is 70

Exception in thread "main" java.lang.ArrayIndexOutOfBoundsException: Index 7 out of bounds for length 7 at Example.main(Example.java:7)

- num.length is 7, so Length : 7 is printed. The loop condition $i \leq \text{num.length}$ allows i to reach 7, but num[7] is invalid (valid indices are 0 to 6). This causes an ArrayIndexOutOfBoundsException at $i = 7$.

ii.

```
public class Example {  
    public static void main(String s[]) {  
        int[] num={10,20,30,40,50,60,70};  
        System.out.println("Length : "+num.length);  
        for(int i=0; i<num.length; i++){  
            System.out.println("Element at index " + i + " is " + num[i]);  
        }  
    }  
}
```

17).

i.

Line 3 (num == arr ??? true) → num = arr assigns the same reference to num. Both point to the same array object in memory, so num == arr returns true (reference comparison).

Line 4 ([I@<hash_code>) → Printing arr directly calls its default toString(), which outputs the array type ([I for int array) and its hash code, not the values.

ii.

System.out.println(num); prints [I@<hash_code1>, the default toString() output for the array object (type [I for int array, same hash code as arr).

Reason: num and arr reference the same array object, so Line 5 outputs the same hash code as Line 4.

Memory:

[10, 20, 30, 40, 50, 60, 70] <-- @<hash_code1>

^	^
arr	num

- Both `arr` and `num` point to the same array.

- Line 5 prints `[I@<hash\_code1>`, matching Line 4.

18).

i.

```
public class Example {
```

```
    public static void main(String args[]){
```

```
        String[] month =
```

```
{"January","February","March","April","June","July","August","September","October","November","December"};
```

```
        String[] arrMonth = new String[month.length];
```

```
        for(int i=0;i<arrMonth.length;i++){
```

```
            arrMonth[i] = month[i];
```

```
        }
```

```
    }
```

```
}
```

ii.

```
public class Example {
```

```

public static void main(String args[]){

    String[] month =
{"January","February","March","April","June","July","August","September","October","November","December"};

    String[] arrMonth = new String[month.length];

    for(int i=0;i<arrMonth.length;i++){

        arrMonth[i] = month[i];

        System.out.println("month == arrMonth ??? "+ (month==arrMonth));

    }

}
}

```

//output→

```

month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false
month == arrMonth ??? false

```

The == operator checks if month and arrMonth refer to the same array object. Since they are two different arrays (different memory locations), month == arrMonth always returns false.

19).

I.

```

import java.util.*;

public class Example {

    public static void main(String args[]){

        Random random = new Random();

        int[] number = new int[50];
    }
}

```

```

for(int i=0;i<number.length;i++){
    number[i] = random.nextInt(100)+1;
}
System.out.print("[");
for(int i=0;i<number.length;i++){
    System.out.print(number[i]+" " );
}
System.out.print("\b\b");

```

```

System.out.println(" ");
System.out.println("New Array - ");
int[] newArray = new int[number.length];
for(int i=0;i<newArray.length;i++){
    newArray[i]=number[i];
}
System.out.print("[");
for(int i=newArray.length-1;i>=0;i--){
    System.out.print(newArray[i]+" " );
}
System.out.print("\b\b");

```

}

}

ii.

```
import java.util.*;
```

```
public class Example {
```

```
    public static void main(String args[]){
```

```
        Random random = new Random();
```

```
        int[] number = new int[50];
```



```

int countPositive = 0;
int countNegative = 0;
for(int i=0;i<number.length;i++){
    number[i] = random.nextInt();
}
System.out.print("[");
for(int i=0;i<number.length;i++){
    System.out.print(number[i]+", " );
}
System.out.print("\b\b]");

System.out.println(" ");
System.out.println("New Array - ");
int[] newArray = new int[number.length];
for(int i=0;i<newArray.length;i++){
    newArray[i]=number[i];
    if(newArray[i]<0){
        countNegative++;
    }else{
        countPositive++;
    }
}

System.out.println("Count Negative numbers - "+countNegative);
System.out.println("Count Positive numbers - "+countPositive);
}
}
20).
i.
import java.util.Scanner;

```

```

public class Example {
    public static void main(String args[]){
        char[] vowels1 ={'A','E','I','O','U'};
        char[] vowels2 ={'a','e','i','o','u'};

        for(int i=0;i<vowels1.length;i++){
            System.out.print(vowels1[i]+" ");

        }
        for(int i=0;i<vowels1.length;i++){
            System.out.print(vowels2[i]+" ");

        }

    }
}

```

ii.

```

public class Example {
    public static void main(String args[]){
        char[] vowels1 ={'A','E','I','O','U'};
        char[] vowels2 ={'a','e','i','o','u'};

        for(int i=0;i<vowels1.length;i++){
            System.out.print(vowels1[i]+" "+vowels2[i]+" ");

        }

    }
}

```

iii.

```

public class Example {

```

```

public static void main(String args[]){
    char[] vowels1 ={'A','E','I','O','U'};
    char[] vowels2 ={'a','e','i','o','u'};

    for(int i=vowels1.length-1;i>=0;i--){
        System.out.print(vowels1[i]+" "+vowels2[i]+" ");
    }

}
}

```

iv.

```

public class Example {
    public static void main(String args[]){
        char[] vowels1 ={'A','E','I','O','U'};
        char[] vowels2 ={'a','e','i','o','u'};

        for(int i=0;i<vowels1.length;i++){
            System.out.print(vowels1[i]+" ");
            System.out.print(vowels2[vowels2.length-1-i]+" ");
        }
    }
}

```

21).

C) dr1[0] = ar1[0];
 E) ar1[0] = (int)dr1[0];

22).

A) It refers to the process of converting a value from a narrower array type to a wider array type.

23).

- A. `x = xr[0];`
- B. `xr[0] = x;`
- E. `dr[0] = xr[0];`
- G. `xr[0] = (int)dr[0];`

24).

i.

Initial salary array: [71.0, 86.0, 54.0, 89.0]

v	v	v	v

increment method: [71, 86, 54, 89] (copies as int)

v	v	v	v

After increment: [5071, 5086, 5054, 5089] (only in method, not in original array)

Final salary array: [71.0, 86.0, 54.0, 89.0] (no change)

71.0 86.0 54.0 89.0

71.0 86.0 54.0 89.0

Java uses pass-by-value for primitive types. The increment method modifies copies of the array elements, not the original salary array, so the array remains unchanged.

ii.

```
import java.util.Scanner;
```

```
public class Example {
```

```
    public static void increment(double[] salary){
```

```
        for(int i=0;i<salary.length;i++){
```

```
            salary[i]+=5000;
```

```
        }
```

```

    }

    public static void main(String args[]){
        double[] salary={71,86,54,89};

        System.out.println(salary [0]+" "+ salary [1]+" "+ salary [2]+" "+ salary [3]);
        increment(salary); //Line 2

        System.out.println(salary [0]+" "+ salary [1]+" "+ salary [2]+" "+ salary [3]);
    }
}

```

iii.

```

import java.util.Scanner;

public class Example {

    public static void increment(double[] salary){
        for(int i=0;i<salary.length;i++){
            salary[i]+=5000;
        }

    }

    public static void main(String args[]){
        double[] salary={71,86,54,89};

        System.out.println("Salary - ");
        System.out.print("[");
        for(double x : salary){
            System.out.print(x+" , ");
        }
        System.out.print("\b\b]");

        System.out.println(" ");
        System.out.println("Increment Salary - ");
        System.out.print("[");
    }
}

```

```

        increment(salary); //Line 2
    for(double x : salary){
        System.out.print(x+" ");
    }
    System.out.print("\b\b");
}
}

```

25).

i.

```

import java.util.Scanner;
import java.util.Arrays;
public class Example {
    public static void main(String[] args) {
        float[] salary = {15000,21000,18500,12275,35550,28760,17175,10500};
        float[] sriLankanSalary = new float[salary.length];

        for(int i=0;i<salary.length;i++){
            salary[i] = salary[i]*366.10f;
        }
        sriLankanSalary = salary;
        System.out.println(Arrays.toString(sriLankanSalary));
    }
}

```

ii.

```

import java.util.Scanner;
import java.util.Arrays;
public class Example {
    public static void main(String[] args) {

```

```
float[] salary = {15000,21000,18500,12275,35550,28760,17175,10500};
```

```
float[] sriLankanSalary = new float[salary.length];
```

```
for(int i=0;i<salary.length;i++){
```

```
    salary[i] = salary[i]*366.10f;
```

```
}
```

```
sriLankanSalary = salary;
```

```
System.out.println(Arrays.toString(sriLankanSalary));
```

```
int countGreaterthan = 0;
```

```
int countLessThan = 0;
```

```
for(int i=0;i<sriLankanSalary.length;i++){
```

```
    if(sriLankanSalary[i]>7500000){
```

```
        countGreaterthan++;
```

```
    }else{
```

```
        countLessThan++;
```

```
    }
```

```
}
```

```
System.out.println("Greater than - "+countGreaterthan);
```

```
System.out.println("Less than - "+countLessThan);
```

```
}
```

```
}
```

iii.

```
import java.util.Scanner;
```

```
import java.util.Arrays;
```

```
public class Example {
```

```
    public static void main(String[] args) {
```

```
        float[] salary = {15000,21000,18500,12275,35550,28760,17175,10500};
```

```
        float[] sriLankanSalary = new float[salary.length];
```

```
for(int i=0;i<salary.length;i++){
    sriLankanSalary[i] = salary[i]*366.10f;
}

System.out.println(Arrays.toString(sriLankanSalary));

int countGreaterthan = 0;
int countLessThan = 0;

for(int i=0;i<sriLankanSalary.length;i++){
    if(sriLankanSalary[i]>7500000){
        countGreaterthan++;
    }else{
        countLessThan++;
    }
}

System.out.println("Greater than - "+countGreaterthan);
System.out.println("Less than - "+countLessThan);

float[] greaterThan = new float[countGreaterthan];
float[] lessThan = new float[countLessThan];

int greaterThanIndex = 0;
int lessThanIndex = 0;
for(int i=0;i<sriLankanSalary.length;i++){
    if(sriLankanSalary[i] > 7500000) {
        greaterThan[greaterThanIndex] = sriLankanSalary[i];
        greaterThanIndex++;
    }
}
```



```

        }else{
            lessThan[lessThanIndex] = sriLankanSalary[i];
            lessThanIndex++;
        }
    }
    System.out.println("Greater than 7500,000");
    System.out.println("=====");
    for(int i=0;i<greaterThan.length;i++){
        System.out.print(greaterThan[i]+",");
    }
    System.out.println(" ");
    System.out.println("Less than 7500,000");
    System.out.println("=====");
    for(int i=0;i<lessThan.length;i++){
        System.out.print(lessThan[i]+",");
    }
}
}

```

26).

i.

```

public class Example {
    public static void main(String[] args) {
        int[] arr = {96,17,54,6,81,47,63,79,32,13};
    }
}

```

ii.

[6, 17, 54, 96, 81, 47, 63, 79, 32, 13]

iii.

→ {96,17,54,6,81,47,63,79,32,13}

→ loop 1 – 96>17

17>54

54>6

T = 96	t = 54
[0] = 17	[2] = 6
[1] = t	[3] = t
ar = {17,96,6,54}	
→ loop 2 – 17>96	96>6 6>54
	T = 96
	[1] = 6
	[2] = t
ar = {17,6,96,54}	
→ loop 3 – 17>6 6>96 96>54	
T = 17	t = 96
[0] = 6	[2] = 54
[1] = t	[3] = t
ar = {6,17,54,96}	
ar = [6, 17, 54, 96, 81, 47, 63, 79, 32, 13]	

iv.

```
import java.util.Scanner;
import java.util.Arrays;
public class Example {
    public static void main(String[] args) {
        int[] arr = {96,17,54,6,81,47,63,79,32,13};
        sort(arr);
    }
    public static void sort(int[] ar){
        for(int j=0; j<3; j++){ // loop1
            if(ar[j]>ar[j+1]){
                int t=ar[j];
                ar[j]=ar[j+1];
                ar[j+1]=t;
            }
        }
        for(int j=0; j<3; j++){ // loop2
            if(ar[j]>ar[j+1]){
                int t=ar[j];
                ar[j]=ar[j+1];
                ar[j+1]=t;
            }
        }
        for(int j=0; j<3; j++){ // loop3
            if(ar[j]>ar[j+1]){
                int t=ar[j];
                ar[j]=ar[j+1];
                ar[j+1]=t;
            }
        }
        System.out.println(Arrays.toString(ar));
        System.out.println("Ascading Order");
        Arrays.sort(ar);
        System.out.println(Arrays.toString(ar));
    }
}
```

27).

28).

```
import java.util.Scanner;

public class Example {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        int[] oldArray = {10,20,30,40};

        int[] newArray = new int[oldArray.length];

        int second = 50;

        System.out.println("Old Array");

        System.out.print("[");

        for(int i=0; i<oldArray.length;i++){

            System.out.print(oldArray[i]+" ");

        }

        System.out.print("\b\b");

        modifyArray(oldArray,newArray,second);

    }

    public static void modifyArray(int[] oldArray,int[] newArray,int second){

        newArray = oldArray;

        System.out.println(" ");

        System.out.println("New Array");

        newArray[newArray.length-2] = second;

        System.out.print("[");

        for(int i=0; i<oldArray.length;i++){

            System.out.print(newArray[i]+" ");

        }

        System.out.print("\b\b");

    }

}
```

```
}
```

29).

```
import java.util.Scanner;
```

```
public class Example {
```

```
    public static void main(String[] args) {
```

```
        Scanner input = new Scanner(System.in);
```

```
        String[] subject = new String[10];
```

```
        subject[0] = "Mathematics";
```

```
        subject[1] = "Science";
```

```
        subject[2] = "English";
```

```
        subject[3] = "Sinhala";
```

```
        subject[4] = "History";
```

```
        subject[5] = "Buddhisum";
```

```
        subject[6] = "Business Studies";
```

```
        subject[7] = "Computer Studies";
```

```
        int subjectCount = 8;
```

```
        int newCount = subjectCount;
```

```
        newCount++;
```

```
        System.out.print("[");
```

```
        for(int i=0; i<newCount; i++){
```

```
            subject[7] = "Music";
```

```
            subject[8] = "Computer Studies";
```

```
            System.out.print(subject[i]+" , ");
```

```
        }
```

```
        System.out.print("\b\b]");
```

```
    }
```

```
}
```

30).

i.

```
class Example {  
    public static void main(String[] args) {  
        String[] months = {"January", "February", "March", "April", "May", "June",  
            "July", "August", "September", "October", "November", "December"};  
        int[] reserves = {2362, 2311, 1917, 1812, 1887, 1854, 1817, 1717, 1779, 1705, 1806, 1896};  
        String[] newMonths = new String[months.length];  
        int[] newReserves = new int[reserves.length];  
  
        System.arraycopy(months, 0, newMonths, 0, months.length);  
        System.arraycopy(reserves, 0, newReserves, 0, reserves.length);  
        System.out.println("New Months: " + java.util.Arrays.toString(newMonths));  
        System.out.println("New Reserves: " + java.util.Arrays.toString(newReserves));  
    }  
}
```

ii.

```
class Example{  
    public static void sortDescending(String[] months, int[] reserves) {  
        int n = reserves.length;  
        for (int i = 0; i < n - 1; i++) {  
            for (int j = 0; j < n - i - 1; j++) {  
                if (reserves[j] < reserves[j + 1]) {  
                    int tempReserve = reserves[j];  
                    reserves[j] = reserves[j + 1];  
                    reserves[j + 1] = tempReserve;  
                    String tempMonth = months[j];  
                    months[j] = months[j + 1];  
                    months[j + 1] = tempMonth;  
                }  
            }  
        }  
    }  
}
```

iii.

```
class Example {
```

```
public static void sortDescending(String[] months, int[] reserves) {
```

```
int n = reserves.length;
```

```
for (int i = 0; i < n - 1; i++) {
```

```
for (int j = 0; j < n - i - 1; j++) {
```

```
if (reserves[j] < reserves[j + 1]) {
```

```
int tempReserve = reserves[j];
```

```
reserves[j] = reserves[j + 1];
```

```
reserves[j + 1] = tempReserve;
```

```
String tempMonth = months[j];
```

```
months[j] = months[j + 1];
```

```
months[j + 1] = tempMonth;
```

}

}

}

}

```
public static void main(String[] args) {
```

```
String[] months = {"January", "February", "March", "April", "May", "June",
```

```
"July", "August", "September", "October", "November", "December"};
```

```
int[] reserves = {2362, 2311, 1917, 1812, 1887, 1854, 1817, 1717, 1779, 1705, 1806, 1896};
```

```

String[] newMonths = new String[months.length];
int[] newReserves = new int[reserves.length];
System.arraycopy(months, 0, newMonths, 0, months.length);
System.arraycopy(reserves, 0, newReserves, 0, reserves.length);

System.out.println("Foreign Exchange Reserves by Month Order:");
for (int i = 0; i < months.length; i++) {
    System.out.println(months[i] + ": " + reserves[i]);
}

sortDescending(newMonths, newReserves);

System.out.println("\nForeign Exchange Reserves in Descending Order:");
for (int i = 0; i < newMonths.length; i++) {
    System.out.println(newMonths[i] + ": " + newReserves[i]);
}
}
}

```

31).

i.

```

import java.util.Scanner;

public class Example {
    public static void main(String args[]){
        String[] name =
{"Sachini","Tharusha","Shehan","Ashini","Sachin","Kasun","Lihini","Hashini","Ashini","Kaveesha"};
        int[] marks = {807,796,728,801,818,746,781,756,801,812};
    }
}

```

```

        for(int i=0; i<name.length; i++){
            System.out.println(name[i]+" - "+marks[i]);
        }
    }
}

```

ii.

```

import java.util.Scanner;

public class Example {

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        String[] name =
{"Sachini","Tharusha","Shehan","Ashini","Sachin","Kasun","Lihini","Hashini","Ashini","Kaveesha"};

        int[] marks = {807,796,728,801,818,746,781,756,801,812};

        for(int i=0; i<name.length; i++){

            System.out.println(name[i]+" - "+marks[i]);

        }

        String name1 = "Amesha";

        int marks1 = 804;

        for(int i=0; i<name.length; i++){

            name[i] = name1;

            marks[i] = marks1;

            System.out.println(name[i]+" - "+marks[i]);

            break;

        }

    }
}

```



```
}
```

iii.

```
import java.util.Scanner;

public class Example {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        String[] name = new String[20];

        int[] marks = new int[20];


        name[0] = "Sachini"; marks[0] = 807;

        name[1] = "Tharusha"; marks[1] = 796;

        name[2] = "Shehan"; marks[2] = 728;

        name[3] = "Ashini"; marks[3] = 801;

        name[4] = "Sachin"; marks[4] = 818;

        name[5] = "Kasun"; marks[5] = 746;

        name[6] = "Lihini"; marks[6] = 781;

        name[7] = "Hashini"; marks[7] = 756;

        name[8] = "Ashini"; marks[8] = 801;

        name[9] = "Kaveesha"; marks[9] = 812;


        int studentCount = 10;

        int newCount = studentCount;

        for (int i = 0; i < newCount; i++) {

            for (int j = i + 1; j < newCount; j++) {

                if (name[i].equals(name[j])) {

                    for (int k = j; k < newCount - 1; k++) {

                        name[k] = name[k + 1];

                        marks[k] = marks[k + 1];
```

```

        }
        newCount--;
        j--;
    }
}
}
studentCount = newCount;

```

```

        String name1 = "Amesha";
int marks1 = 804;
name[studentCount] = name1;
marks[studentCount] = marks1;
studentCount++;
for (int i = 0; i < studentCount; i++) {
    System.out.println(name[i] + " - " + marks[i]);
}
}
}

```

iv.

```

import java.util.Scanner;

public class Example {

    public static void main(String[] args) {

        Scanner input = new Scanner(System.in);

        String[] name = new String[20];
        int[] marks = new int[20];

        name[0] = "Sachini"; marks[0] = 807;
        name[1] = "Tharusha"; marks[1] = 796;
        name[2] = "Shehan"; marks[2] = 728;
    }
}

```

```
name[3] = "Ashini"; marks[3] = 801;
name[4] = "Sachin"; marks[4] = 818;
name[5] = "Kasun"; marks[5] = 746;
name[6] = "Lihini"; marks[6] = 781;
name[7] = "Hashini"; marks[7] = 756;
name[8] = "Ashini"; marks[8] = 801;
name[9] = "Kaveesha"; marks[9] = 812;
```

```
int studentCount = 10;

    int newCount = studentCount;
for (int i = 0; i < newCount; i++) {
    for (int j = i + 1; j < newCount; j++) {
        if (name[i].equals(name[j])) {

            for (int k = j; k < newCount - 1; k++) {
                name[k] = name[k + 1];
                marks[k] = marks[k + 1];
            }
            newCount--;
            j--;
        }
    }
}

studentCount = newCount;
```

```
    String name1 = "Amesha";
int marks1 = 804;
name[studentCount] = name1;
marks[studentCount] = marks1;
```

```

studentCount++;
for(int i=0;i<studentCount;i++){
    for(int j=i+1;j<studentCount;j++){
        if(marks[i]<marks[j]){
            int tempM = marks[i];
            marks[i] = marks[j];
            marks[j] = tempM;

            String tempN = name[i];
            name[i] = name[j];
            name[j] = tempN;
        }
    }
    System.out.println(name[i]+" - "+marks[i]);
}
}
}

```

32).

```

public class Example {
    public static void main(String[] args) {
        int[] arr = {10,20,30,40,50,60,70};
        rearrangeArray(arr);
    }
    public static void rearrangeArray(int[] arr){
        for(int i=1;i<arr.length;i++){
            System.out.print(arr[i]+",");
            if(arr[i]==70){
                System.out.print(arr[0]);
            }
        }
    }
}

```

```

        }
    }
}

```

33).

```

public class Example {
    public static void main(String[] args) {
        int[] binary = {1,0};
        for(int i=0;i<binary.length;i++){
            int temp1 = 0;
            int temp2 = 0;
            if(binary[i]!=0){
                temp1 += binary[i];
                binary[i] = 0;
                temp2 += temp1;
                System.out.print(binary[i]+",");
                System.out.print(temp1);
                //break;
            }else{
                System.out.print(binary[i]+",");
            }
        }
    }
}

```

34).

```

public class Example {
    public static void main(String[] args) {
        String[] countries = {
            "Switzerland(2)", "Lvory Coast(1)", "Jamaica(2)", "United States(1)",
            "United States(2)", "Great Britain", "Lvory Coast(2)", "Jamaica(3)", "Switzerland(1)"
        }
    }
}

```

```
};
```

```
float[] times = {
```

```
    11.11f, 10.86f, 10.76f, 10.98f, 11.11f, 11.05f, 11.14f, 10.61f, 11.06f
```

```
};
```

```
    for (int i = 0; i < times.length - 1; i++) {
```

```
    for (int j = i + 1; j < times.length; j++) {
```

```
        if (times[j] < times[i]) {
```

```
            float tempTime = times[i];
```

```
            times[i] = times[j];
```

```
            times[j] = tempTime;
```

```
        String tempCountry = countries[i];
```

```
        countries[i] = countries[j];
```

```
        countries[j] = tempCountry;
```

```
    }
```

```
}
```

```
}
```

```
    System.out.println("Gold medal winner - ");
```

```
    System.out.print("\t"+countries[0]+" - ");
```

```
    System.out.println(times[0] + "s");
```

```
    System.out.println("New Gold medal winner - ");
```

```
    System.out.print("\t"+countries[0]+" - ");
```

```
    System.out.print(times[0]+"s");
```

```
    System.out.println(" ");
```

```

float[] newArray = new float[times.length];
newArray = times;
String[] update = new String[countries.length];
update = countries;
System.out.println("Gold medal in 2021 and rearrange the list in ascending order...");
for(int i=1;i<times.length;i++){
System.out.println(countries[i]+" - "+times[i]+"s");

}
}
}

```

35).

```

class Example{
    public static void main(String args[]){
        int[] ar={10,20,30,40,50};
        int[] br={10,20,10,40,50,60};
        int[] cr={30,30,30,30};
        int[] dr={40,10,50,20,30};
        int[] er={10,20,30,40,50};
        int[] fr={10,20,30,40};

        findDuplicate(ar); // no duplicate
        findDuplicate(br); // duplicate "10"
        findDuplicate(cr); // dupliacte "30"
        findDuplicate(dr); // no duplicate
        findDuplicate(er); // no duplicate
        findDuplicate(fr); // no duplicate

        System.out.println("ar==br : "+Arrays.equals(ar,br)); //false
    }
}

```

```

        System.out.println("ar==cr : "+Arrays.equals(ar,cr)); //false
        System.out.println("ar==dr : "+Arrays.equals(ar,dr)); //false
        System.out.println("ar==er : "+Arrays.equals(ar,er)); //true
        System.out.println("ar==fr : "+Arrays.equals(ar,fr)); //false*/
    }

    public static int findDuplicate(int[] array) {
        for (int i = 0; i < array.length; i++) {
            for (int j = i + 1; j < array.length; j++) {
                if (array[i] == array[j]) {
                    System.out.println("Duplicate found: " + array[i]);
                    return 1;
                }
            }
        }
        System.out.println("No duplicates found.");
        return 2;
    }
}

```

36).

```

import java.util.*;

class Example{

    public static void main(String args[]){

        Scanner input = new Scanner(System.in);

        String[] cricket =
{"Shanaka","Dinesh","Amesh","Lahiru","Dilshan","Hashan","Wenura","Randika","Dasun","Thisara","Kasun","Nisal","Praveen"};

        String[] football =
{"Anjana","Chamara","Kamal","Dhanuka","Shanaka","Kasun","Dineth","Sahan","Wenura","Amesh","Mahesh","Randika","Thisara"};

```



```

System.out.println("Name list of students playing both cricket and football :");
for(int i=0; i<cricket.length;i++){
    for(int j=0;j<football.length;j++){
        if(cricket[i].equals(football[j])){
            System.out.print(cricket[i]+",");
        }
    }
}

System.out.println(" ");
System.out.println("Name list of student playing cricket or football : ");
Set<String> uniqName = new HashSet<>();
uniqName.addAll(Arrays.asList(cricket));
uniqName.addAll(Arrays.asList(football));

for(String word : uniqName){
    System.out.print(word+",");
}

}

```

37).

```

import java.util.*;

class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        System.out.print("Input string - ");
        String string = input.nextLine();

        char[] ch = string.toCharArray();
        System.out.print(Arrays.toString(ch));
    }
}

```

```

    }
}
38).
39).
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        ArrayList<Integer> registrationNum = new ArrayList<>();
        ArrayList<String> name = new ArrayList<>();
        String again;
        while(true){
            System.out.print("Input registration number - ");
            int number = input.nextInt();
            input.nextLine();

            if(registrationNum.contains(number)){
                System.out.println("Duplicate Admission number ... ");
                System.out.print("Enter another admission number(yes/no) - ");
                again = input.nextLine();
                if(again.equalsIgnoreCase("yes")){
                    continue;
                }else{
                    break;
                }
            }

            System.out.print("Input name - ");
            String rname = input.nextLine();

```

```

        registrationNum.add(number);
        name.add(rname);
    }
    System.out.println(" ");
    System.out.println("!!!!Student Registration System!!!!\n");
    for(int i=0; i<registrationNum.size();i++){
        System.out.println(registrationNum.get(i)+","+name.get(i));
    }
}
}
40).
import java.util.*;
class Example{
    public static void main(String args[]){
        Scanner input = new Scanner(System.in);
        ArrayList<Integer> clothingId = new ArrayList<>();
        L1:while(true){
            System.out.print("Input ID - ");
            int id = input.nextInt();

            if(id<0){
                System.out.println("You enter -1");

                break L1;
            }
            if(clothingId.contains(id)){
                System.out.println("Your number is duplicate number");
            }else{

```

```
        clothingId.add(id);
    }
}
System.out.print("Clothing ID - ");
for(int i=0; i<clothingId.size();i++){
    System.out.print(clothingId.get(i)+",");
}
}
}
```